

**Ryan Miller,
Assistant Professor of Statistics**

Grinnell College

CONTACT INFORMATION	Grinnell College, 1115 8th Avenue, Grinnell, IA, 50112	phone: (641) 269-4017 email: millerry@grinnell.edu website: remiller1450.github.io
EDUCATION	University of Iowa , Iowa City, IA Ph.D., Biostatistics, August 2018 • Thesis: <i>Marginal False Discovery Rate Approaches to Inference on Penalized Likelihood Models</i> • Advisor: Patrick Breheny, Ph.D M.S., Biostatistics, May 2015 • Preceptorship: <i>Analyzing the Geographic Dispersion of Accidents Involving Animal Strikes in Iowa using a Bayesian Spatio-temporal Model</i> • Advisor: Jake Oleson, Ph.D Augustana College , Rock Island, IL B.A., Mathematics and Mathematics Education (Double Major), May 2011 • Graduated <i>Cum Laude</i> • Advisor: Stacey Rodman, Ph.D	
PROFESSIONAL EXPERIENCE	Assistant Professor of Statistics Department of Statistics Grinnell College	2022 to present
	Assistant Professor of Statistics (tenure track) Department of Mathematics Xavier University	2020 to 2022
	Visiting Assistant Professor of Statistics Department of Mathematics and Statistics Grinnell College	2018 to 2020
	Research Assistant Iowa Public Policy Center, University of Iowa, Role: Statistical support on projects involving Iowa Department of Transportation crash records, state driver licensure databases, and the State Innovation Model for Iowa healthcare	2014 to 2018
TEACHING EXPERIENCE	Grinnell College, Department of Statistics STA 370 - Introduction to Machine Learning (4) Spring 2023, Spring 2024, Fall 2025, Fall 2026 STA 330 - Applied Data Science (3) Spring 2020, Spring 2023, Spring 2025 STA 230 - Introduction to Data Science (7) Spring 2019 (2), Fall 2019, Fall 2022 (2), Spring 2023, Fall 2023 STA 209 - Applied Statistics (11) Fall 2018 (2), Spring 2019, Fall 2019 (2), Spring 2020, Spring 2024, Spring 2025 (2), Fall 2025, Fall 2026	

TUT 100 - Tutorial (1)
Fall 2023

Xavier University, Department of Mathematics

MATH 257 - Data Modeling (1)
Spring 2021
MATH 256 - Introduction to Probability and Statistics (2)
Fall 2021, Fall 2021
MATH 156 - General Statistics (4)
Spring 2022, Spring 2021 (2), Fall 2020
MATH 146 - Introductory Biostatistics (4)
Spring 2022 (2), Fall 2021 (2)

University of Iowa, Department of Biostatistics

BIOS 4120 - Introduction to Biostatistics
Summer 2014 (Instructor), Fall 2014 (TA), Spring 2014 (TA)

Pawnee High School (IL)

Calculus I, Algebra II w/ Trig, Algebra I, Advanced Algebra 2011-2012

PEER-REVIEWED
PUBLICATIONS

9. **Miller, R.**, Chestovich, A*, Satav, A*, Brown, T, “Detection of Alcohol-Impaired Driving using Eye-Tracking and Vehicle Data via Machine Learning” *Journal of Safety Research* 2026 (accepted, to appear)
8. **Miller, R.**, Shaikh, A*, Mukherjee, T*, Schwarz, C, Brown, T “Clustering Roadway Departures from Impaired Driving Studies using Derived Features and Shape-Based Methods” *Transportation Research Record* 2026
<https://journals.sagepub.com/doi/10.1177/03611981261441282>
7. **Miller, R.**, Carlton, B*, Brown, T “Detecting Impaired Driving: Vehicle Inputs with the Greatest Influence on Accurate Identification” *Transportation Research Record* 2025
<https://journals.sagepub.com/doi/10.1177/03611981251348460>
6. **Miller, R.**, Brown, T, Schmitt, R, Gaffney, G, Milavetz, G “Predicting changes in driving performance in individuals who use cannabis following acute use based on self-reported readiness to drive” *Accident Analysis and Prevention*. 2023
<https://pubmed.ncbi.nlm.nih.gov/37984112/>
5. **Miller, R.**, Breheny, P “Feature-specific inference for penalized regression using local false discovery rates” *Statistics in Medicine*. 2023
<https://onlinelibrary.wiley.com/doi/10.1002/sim.9678>
4. **Miller, R.**, Brown, T, Wrobel, J, Kosnett, M, Brooks-Russell, A “Influence of cannabis use history on the impact of acute cannabis smoking on simulated driving performance during a distraction task” *Traffic Injury Prevention*. 2022
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10108575/>
3. Vo, M*, Kuo-Esser, L*, Dominguez, M*, Barta, H*, Graber, M*, Rausenberger, A*, **Miller, R.**, Sommer, N, Escorcía, W “Photo Phenosizer, a rapid machine learning-based method to measure cell dimensions in fission yeast” *microPublication Biology*. 2022
<https://micropublication.org/journals/biology/micropub-biology-000620>

2. **Miller, R**, Brown, T, Lee, S*, Tibrewal, I*, Milavetz, G, Gaffney, G, Hartman, R, D Gorelick, R Compton, Huestis, M “Impact of cannabis and low alcohol concentration on divided attention tasks during driving” *Traffic Injury Prevention*. 2020
<https://www.tandfonline.com/doi/full/10.1080/15389588.2020.1814956>
 1. **Miller, R**, Breheny, P “Marginal false discovery rate control for likelihood-based penalized regression models” *Biometrical Journal*. 2019
<https://onlinelibrary.wiley.com/doi/abs/10.1002/bimj.201800138>
- * indicates undergraduate student co-author

MANUSCRIPTS IN-PROGRESS

2. **Miller, R**, Breheny, P “Marginal False Discovery Rates for Inference on Nonlinear Effects and Categorical Predictors using Group Penalized Models” - submitted in April 2026, currently under review
 1. **Miller, R**, Satav, A*, Safiyazov, A* “FDF-ROCKET: A False Discovery Rate Filtering Approach for Explainable Time Series Classification” - submitted in August 2026, currently under review
- * indicates undergraduate student co-author

REFEREED PRESENTATIONS

9. **Miller, R**, Safiyazov, A*, Satav, A*, Brown, T, “Interpretable Gaze-Based Detection of Alcohol-Impaired Driving: Can Engineered Features Achieve Parity with Black-Box Models?” Accepted and forthcoming at *Transportation Research Board Annual Meeting*. January 2027
8. **Miller, R**, Chestovich, A*, Satav, A*, Brown, T, “Detection of Alcohol-Impaired Driving using Eye-Tracking and Vehicle Data via Machine Learning” *Transportation Research Board Annual Meeting, Section on Emerging Technologies and Driver Behavior*. January 2026
7. **Miller, R**, Shaikh, A*, Mukherjee, T*, Schwarz, C, Brown, T, “Clustering Roadway Departures from Impaired Driving Studies using Derived Features and Shape-Based Methods” *Transportation Research Board Annual Meeting, Section on Behavioral Safety*. January 2026
6. **Miller, R**, Agusti Mazza, I*, Brown, T, “Data Augmentation using a Semi-Reflective Generative Model to Improve Automated Detection of Impairment” *Transportation Research Board Annual Meeting, Section on Artificial Intelligence and Advanced Computing*. January 2025
5. **Miller, R**, Carlton B*, Brown, T, “Detecting Impaired Driving: Vehicle Inputs with the Greatest Influence on Accurate Identification” *Transportation Research Board Annual Meeting, Section on Impairment in Transportation*. January 2025
4. **Miller, R**, Hodson, S*, Le, T*, Schmitt, R, Milavetz, G, Brown, T, “Detection of Cannabis Impaired Driving from Vehicle-based Inputs using Machine Learning Methods” *Transportation Research Board Annual Meeting, Section on Emerging Research in Impaired Driving*. January 8, 2024
3. **Miller, R**, Brown, T, Schmitt, R, Gaffney, G, Milavetz, G “Predicting changes in driving performance in individuals who use cannabis following acute use based on self-reported readiness to drive” *Association for the Advancement of Automotive Medicine, 67th annual scientific conference*. October 5, 2023

2. **Miller, R**, Brown, T, Wrobel, J, Kosnett, M, Brooks-Russell, A “Influence of cannabis use history on the impact of acute cannabis smoking on simulated driving performance during a distraction task” *Association for the Advancement of Automotive Medicine, 66th annual scientific conference*. October 12, 2022
1. **Miller, R**, Brown, T, Lee, S*, Tibrewal, I*, Milavetz, G, Gaffney, G, Hartman, Hartman R, D Gorelick, R Compton, Huestis, M “Secondary Task Performance While Driving: The Impacts of cannabis and low alcohol concentration on divided attention tasks during driving” *Association for the Advancement of Automotive Medicine, 64th annual scientific conference*. October 16, 2020

* indicates undergraduate student co-author

MENTORING

Student Research (Mentored Advanced Projects) - 19 Students:

- Safiyazov, A, Satav, A, “Interpretable Gaze-Based Detection of Alcohol-Impaired Driving: Can Engineered Features Achieve Parity with Black-Box Models?”
 - Accepted for presentation at the 2027 *Transportation Research Board Annual Meeting (section information forthcoming)* in Washington DC
- Olsen, J, Kafumbe, J, “Changes in Driver Performance and Eye Behavior in Response to Alcohol, Divided Attention, and Self-Reported Fatigue”
- Shaikh, A, Mukherjee, T “Clustering Roadway Departures from Impaired Driving Studies using Derived Features and Shape-Based Methods”
 - Accepted for poster presentation at the 2026 *Transportation Research Board Annual Meeting (Section on Behavioral Safety)* in Washington DC
 - Accepted for publication in *Transportation Research Record*, 2026
- Chestovich, A, Satav, A “Detection of Alcohol-Impaired Driving using Eye-Tracking and Vehicle Data via Machine Learning”
 - Accepted for oral presentation at the 2026 *Transportation Research Board Annual Meeting (Section on Emerging Technologies and Driver Behavior)* in Washington DC
 - Accepted for publication in *Journal of Safety Research*, 2026
- Carlton, B, Agusti Mazza, I, and Nguyen, M “Data augmentation and feature selection for improved automated detection of drug-impaired driving using machine learning”
 - Accepted for oral presentation at the 2025 *Transportation Research Board Annual Meeting (Artificial Intelligence and Advanced Computing Applications)* in Washington DC
 - Accepted for publication in *Transportation Research Record*, 2025
- Hodson, S, and Le, T “Detection of cannabis impaired driving from vehicle-based inputs using machine learning methods”
 - Accepted for oral presentation at the 2024 *Transportation Research Board Annual Meeting (Emerging Research in Impaired Driving)* in Washington DC
- Krysan, M, and Worth, M “Analysis of driving performance using semi-reflective autoregressive time-series models” (co-mentored)
 - Accepted for poster presentation at the 2023 *Midstates Consortium for Math and Science* at University of Chicago
- Li, L “Facial emotion expression recognition by college students with diverse lived experiences” (co-mentored 2-credit project continuation)
 - Accepted for poster presentation at the 2023 *Midwestern Psychological Association Meeting* in Chicago
- Jindal, T, and Luo, Y “A latent variable approach for analyzing collision scenarios with application to driving simulator time-series data”
 - Accepted for poster presentation at the 2019 *Midstates Consortium for Math and Science* at University of Chicago

- Lee, S, and Tibrewal, I “Evaluating the impact of cannabis, with and without alcohol, on secondary tasks performed while driving”
 - Accepted for oral presentation at the 2019 *Midstates Consortium for Math and Science* at University of Chicago
 - Accepted for oral presentation at the 2020 *Association for the Advancement of Automotive Medicine* annual conference in Portland Oregon
 - Accepted for publication in *Traffic Injury Prevention*, 2020

Award Winning Student Projects:

- Nguyen, L and Duong, D “Spatial and Demographic Effects on Theft Distribution Across Los Angeles” (Spring 2025)
 - 1st place in the [Spring 2025 USCLAP Introductory Statistics paper competition](#)
 - Presented at the [2025 Electronic Undergraduate Statistics Research Conference](#)
- Vu, A, Le Q, and Nguyen, D “Identification Of Effective Biomarkers In Predicting The Survival Of Patients With Severe Sepsis And Septic Shock” (Fall 2022)
 - 2nd place in the [Fall 2022 USCLAP Intermediate Statistics paper competition](#)
- Uzzi, MJ, and Li, L “An Evaluation of the Factors Influencing College Graduation Rate” (Spring 2020)
 - 2nd place in the [Spring 2020 USCLAP Introductory Statistics paper competition](#)
 - Presented at the [2020 Electronic Undergraduate Statistics Research Conference](#)
- Carroll-Townsend, D “Factors on Receiving the Flu Vaccine” (Spring 2020)
 - Honorable mention in the [June 2020 USCLAP Introductory Statistics paper competition](#)

Senior Project Faculty Mentor (Xavier University):

- Filipiak, E “An improved analysis of performance degradation during divided attention tasks performed while driving” (Xavier University, 2021-22)
- Siemer, K “Factor analysis of a novel school climate survey instrument” (Xavier University, 2021-22)

FUNDING

- 1N5 School Climate Instrument Validation (co-investigator with Dr. Victor Ronis-Tobin) - \$10,000 (2020-22)

MISC. HONORS,
RECOGNITIONS

- Project NExT Brown Cohort 2020-2021
- William R. Clarke Graduate Teaching Award recipient, University of Iowa 2018
- Outstanding Teaching Assistant nominee, University of Iowa 2013-14
- Pi Mu Epsilon honor society inductee, Augustana College chapter 2010
- Four-year Varsity Letterman, Cross Country, Track and Field, Augustana College 2007-2011

PROFESSIONAL
SERVICE

Referee for the following journals (16 total articles reviewed):

- Biostatistics (1)
- Statistics in Medicine (1)
- Journal of Statistical Computation and Simulation (1)
- Computational Statistics and Data Analysis (1)
- Accident Analysis and Prevention (3)
- Traffic Injury Prevention (2)
- Transportation Research Record (2)
- Frontiers in Sports and Active Living (1)
- Scientific Reports (1)
- PLOS One (1)

- Drug and Alcohol Review (2)

AP Statistics Exam Reading:

- Exam Reader (5 years) - 2021-2025
- Table Leader (1 year) - 2026

INSTITUTIONAL SERVICE

Member of the following college-wide committees:

- Science Division Representative to the Committee of Academic Standing (Fall 2026-present)
- Health Professions Advisory Committee (AY 2023-24, 24-25, 25-26)
- Academic Honesty Subcommittee Member for the Committee of Academic Standing (Fall 2025)

Other institutional service:

- Grinnell Science Project (GSP) lab instructor (Fall 2025, Fall 2026)
- International Pre Orientation Program (IPOP) faculty panelist (Fall 2026)

DEPARTMENTAL SERVICE

Participated in the following faculty searches:

- Tenure-track faculty search committee (Fall 2022, Fall 2025)
- Term faculty search committee (Spring 2024)

REFERENCES

Available upon request

Last updated: September, 2026